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to the mean position. The displacement of the wave at an instant t is

$$y = y(x,t) = A \sin(k x - \omega t)$$

At the next instant of time $t' = t + \Delta t$ the position of the point P is $x' = x + \Delta x$. Hence, the displacement of the wave at this instant is

$$y = y(x', t') = y(x + \Delta x, t + \Delta t)$$
$$= A \sin[k (x + \Delta x) - \omega (t + \Delta t)] \quad (11.44)$$

Since the shape of the wave remains the same, this means that the phase of the wave remains constant (i.e., the y- displacement of the point is a constant). Therefore, equating equation (11.42) and equation (11.44), we get

$y(x',t') = y(x,t)$, which implies

$$A \sin[k (x + \Delta x) - \omega (t + \Delta t)] = A \sin(k x - \omega t)$$

Or

$$k (x + \Delta x) - \omega (t + \Delta t) = k x - \omega t = \text{constant} \quad (11.45)$$

On simplification of equation (11.45), we get

$$v = \frac{\Delta x}{\Delta t} = \frac{\omega}{k} = v_p \quad (11.46)$$

where v_p is called wave velocity or phase velocity.

By expressing the angular frequency and wave number in terms of frequency and wave length, we obtain

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$k = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k} = \lambda f$$

EXAMPLE 11.15

A mobile phone tower transmits a wave signal of frequency 900MHz. Calculate the length of the waves transmitted from the mobile phone tower.

Solution

Frequency, $f = 900\,MHz = 900 \times 10^6\,Hz$

The speed of wave is $c = 3 \times 10^8 \text{m s}^{-1}$

$$\lambda = \frac{v}{f} = \frac{3 \times 10^8}{900 \times 10^6} = 0.33m$$

11.7 SUPERPOSITION PRINCIPLE

When a jerk is given to a stretched string which is tied at one end, a wave pulse is produced and the pulse travels along the string. Suppose two persons holding the stretched string on either side give a jerk simultaneously, then these two wave pulses move towards each other, meet at some point and move away from each other with their original identity. Their behaviour is very different only at the crossing/meeting points; this behaviour depends on whether the two pulses have the same or different shape as shown in Figure 11.26.

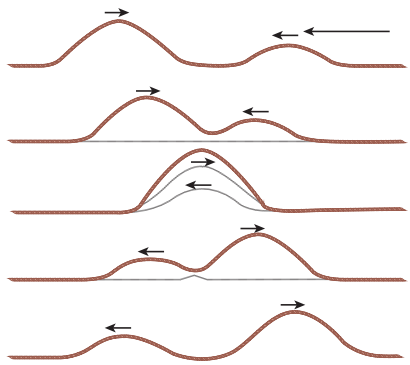


Figure 11.26 Superposition of two waves